Screening for modifiers of genomic imprinting in human blood reveals a role for several genetic and non-genetic factors

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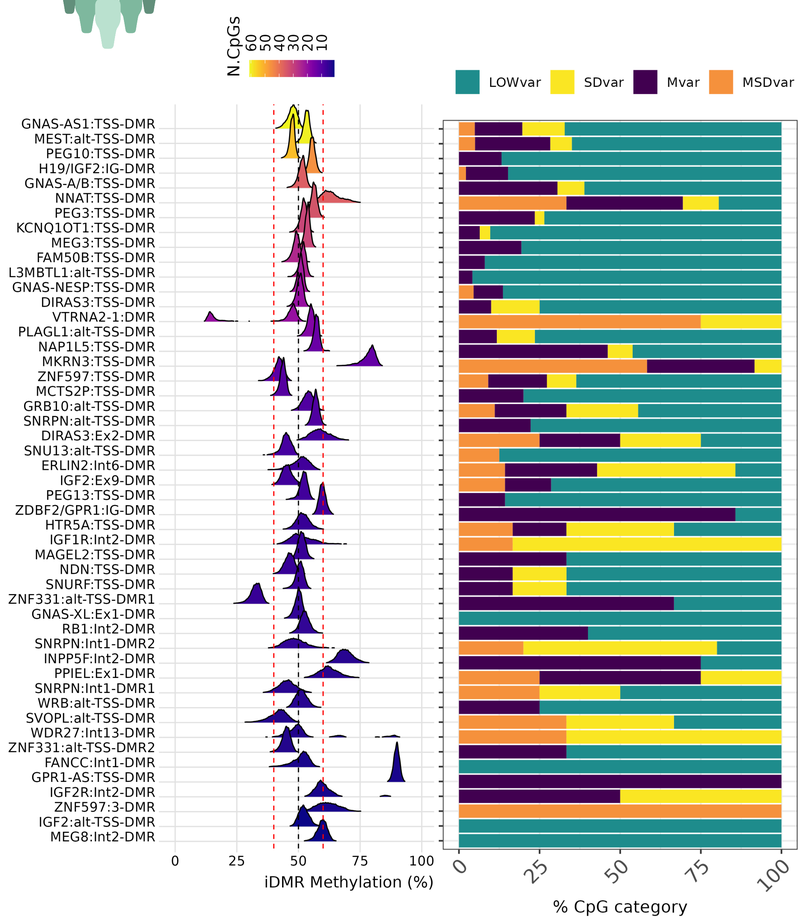
Introduction

Genomic imprinting is required for normal development and abnormal methylation of the differentially methylated regions (iDMRs) controlling the parent of origin-dependent expression of the imprinted genes has been found in constitutional disorders affecting growth, metabolism and neurobehavior and in cancer, yet in most of these cases the cause of the imprinting abnormalities is unknown. Also, these studies have generally been performed on a limited number of CpGs, and a systematic investigation of iDMR methylation in the general population is lacking.

Methylation profiles of the iDMRs in the general population.



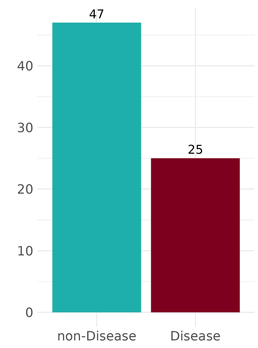
We determined the methylation level of 49 human iDMRs in the peripheral blood leukocyte DNA of a normal population by retrieving and reanalyzing the methylation array datasets of 2664 individuals. We calculated the standard deviation (sd) and the mean methylation level of each iDMR CpG.

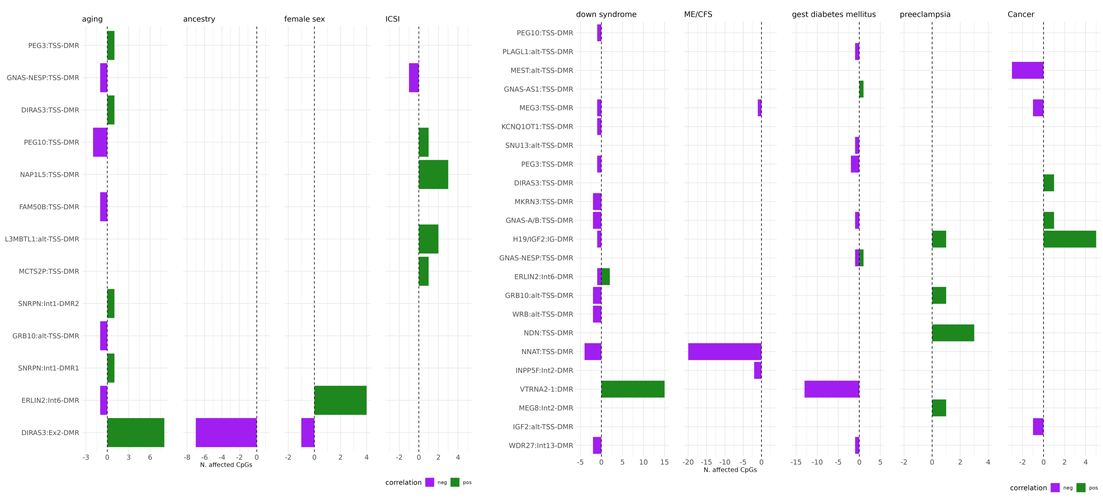
- 117 **SDvar** (sd>0.05)
- 194 **Mvar** (mean methylation outside the [40%, 60%] range)
- 58 **MSDvar**
- 457 **LOWvar**

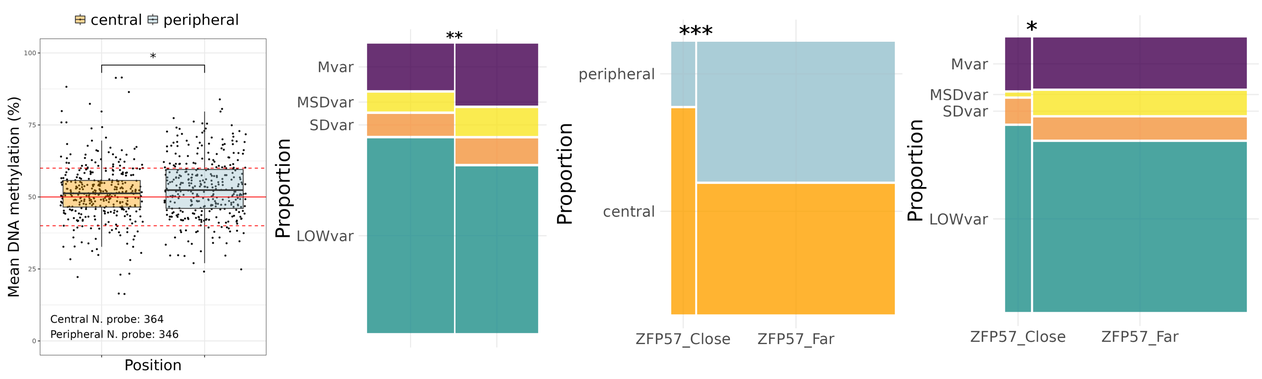
Association of blood iDMR methylation with complex traits.

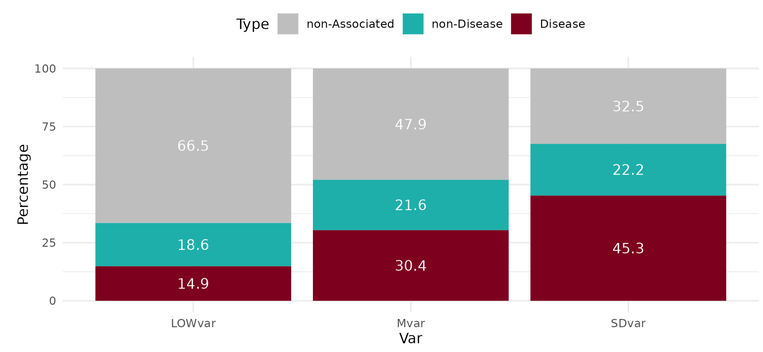
We retrieved the information deposited in the EWAS Data Hub and Atlas for the CpGs of the iDMRs. These data included high-quality associations of about 700 complex traits with CpG methylation in 3000 cohorts and 200 tissues/cells. we distinguished the traits into two categories:

- **non-disease** environmental factors and phenotypic-behavioral traits)
- **disease** traits (neoplastic and non-neoplastic diseases).



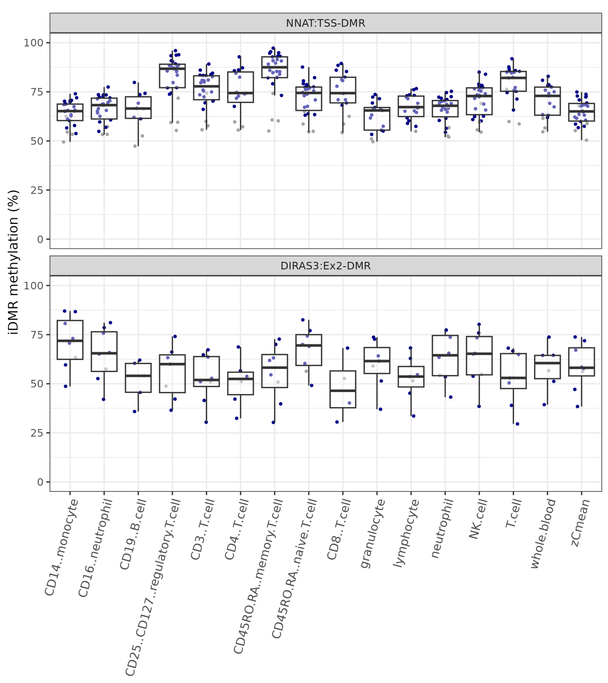


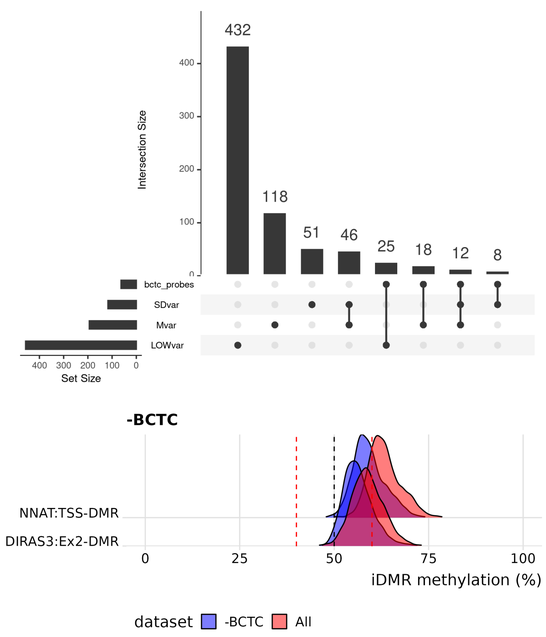




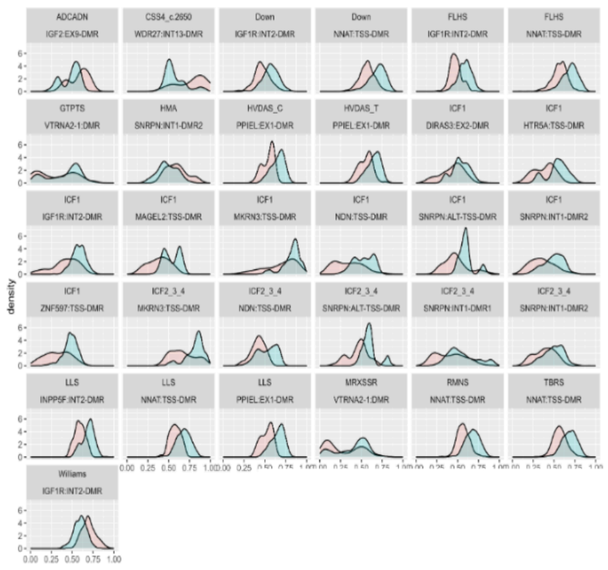
Blood cell type-specific methylation of the iDMRs

We looked at the methylation level of the iDMR CpGs in the datasets of different blood cell types in the EWAS datahub. We found 63 iDMR CpGs whose methylation level had sd>0.05 across all the blood cell types. The BCTC effect was more substantial on NNAT:TSS and DIRAS3:Ex2 with 22 and 7 affected CpGs, respectively,

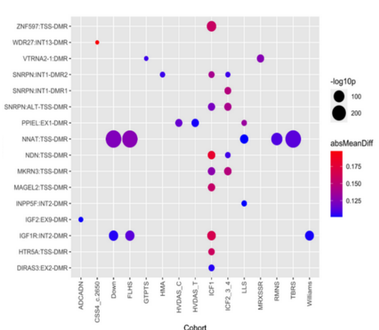


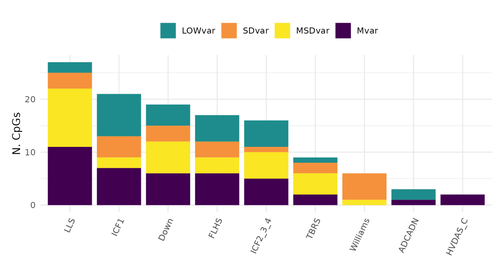


iDMR methylation in developmental disorders.



To identify genes that possibly affect genomic imprinting in humans, a screen for modifiers of iDMR methylation was conducted in samples of patients affected by 56 DDs that display mutations in the proteins of the epigenetic machinery and exhibit genome-wide DNA methylation episignatures within the EpiSign Knowledge Database (EKD)





Conclusion

These findings identify several genetic and non-genetic factors controlling genomic imprinting maintenance in humans, which may have a role in the aetiology of the diseases associated with imprinting abnormalities and should be taken into consideration in all cases where genomic imprinting is investigated for diagnostic purposes.